

3D analysis of the body center of mass in rock climbing

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Abstract

The purpose of the present study was to search for common patterns and for differences in climbing strategies in a group of recreational climbers. Twelve participants were involved in the study. Each participant climbed a simple indoor route consisting of a 3 m horizontal shift followed by a 3 m ascent for five times. Climbers could choose their own style, their preferred speed and holds. Their motion was recorded through motion capture based on passive markers. Results suggested that two main climbing strategies were used: the first preferring agility over force and the second preferring force over agility. We also found that our best climbers tried to minimize power during all trials.

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1. Introduction

Rock climbing has increased in popularity in the last decade both as a recreational physical activity and as a competitive sport. For the Italian Alpine Club the growth of this

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sport was more than 200% in less than 10 years and 118 climbing instructors are teaching about 3 courses/year (CAI website: www.cai.it); the Italian Climbing Federation (FASI) has more than 8000 athletes with a growth of 10% every year (FASI website: www.feder-climb.it). Although it is difficult to precisely quantify the number of climbers worldwide, it seems reasonable to think that it is in the million's range: a quick search on the World Wide Web with the keyword 'rock climbing' scored more than 1450000 web pages and websites from all around the world (Google website, 2007: www.google.com).

Along with this quick development, questions have risen about safety and equipment, but also about biomechanics, movement strategies, and learning capacity.

In reviewing the scientific literature, we found only a few authors dealing with the analysis of rock climbing movements: Cordier and colleagues (Cordier, France, Bolon, & Pailhous, 1993; Cordier, France, Pailhous, & Bolon, 1994) analyzed the learning strategy of a complex motor task using entropy as a global variable; Bourdin, Teasdale, and Nougier (1996) worked on the organization of reaching and grasping movements due to the strong postural constraints induced by rock climbing; Quaine dealt with equilibrium tasks (Quaine, 1997, 1999) and force distribution on the holds (Quaine, Martin, & Blanchi, 1997), but only for a single movement; a few authors (for example, Klauser et al., 2002; Peters, 2001; Rohrbough, Mudge, & Schilling, 2000) focused their studies on the most frequent injuries in climbers; Boschker and Bakker (2002) recorded climbers with a video camera to analyze observational learning strategies; Noè (2006) analyzed the effects of supporting wall inclination on the anticipatory postural adjustments in rock climbing tasks. All studies cited, except Boschker and Bakker's, were performed on elite climbers.

In the present study, we searched for common patterns and for different climbing strategies in a group of non-professional climbers.

2. Materials and methods

2.1. Data collection

Twelve recreational climbers participated in the experiment. Their mean age was 30.6 years (range: 16–49 years); their mean height was 1.74 m (SD 0.11 m) and their mean weight was 64 kg (SD 14.96 kg). After the experimental protocol was fully explained to them, they filled out a questionnaire with their anthropometric data and their climbing experience: in particular we were interested in how long they had been practicing rock climbing (1 month to 25 years), in their best lifetime performance in terms of route difficulty (4b to 7b, French rating scale (Union Internationale des Associations d'Alpinisme, UIAA website: www.uiua.ch; The free dictionary encyclopedia: <http://encyclopedia.thefreedictionary.com/Rock+climbing+%2fggrading>)), and in how many times a week they were training. This last information was used to classify the participants as recreational (not competitive) climbers: all participants declared to train once or twice a week. Moreover, none of them took part in competitions or professional events and none followed a rigorous training program.

Trials were performed on a vertical wall with several holds of various shapes, orientations, and dimensions. The mean horizontal and vertical distance between the holds was about 0.4 m, thus all participants could reach the holds easily independent of their anthropometric characteristics. The variety and dimension of the holds simulated a real climbing environment; the difficulty of the route was approximately 4b (French rating scale). The

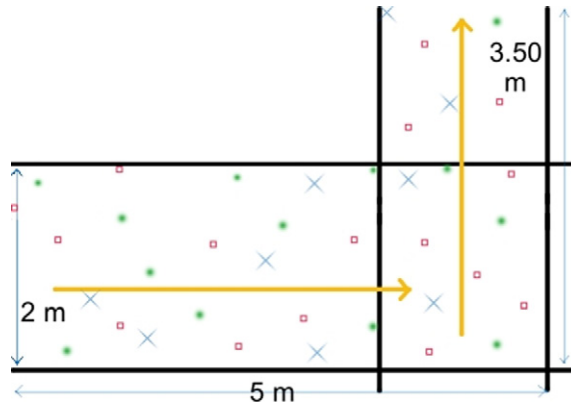


Fig. 1. Schematic view of the climbing wall: the climbing zone defined for the experimental session is delimited by the black lines; squares, crosses and circles represent the different types of holds and their position.

climbing frame was the same for all participants. A schematic view of the climbing wall and of the hold positions is depicted in Fig. 1, where the climbing zone defined for the experimental session is delimited by black lines and the climbing direction is indicated by the orange arrows.

Each participant was asked to climb a 3 m horizontal shift followed by a 3 m ascent, using his preferred technique and climbing at his own speed. Climbers were not forced to grasp any particular hold, but they were asked to move only in the climbing zone delimited by the black lines (see Fig. 1). Each climber had to repeat the task five times; a pause of 30 s was allowed between repetitions. All the climbers wore stretch clothes, and their own climbing harness and climbing shoes, while being secured top-rope during the whole experimental session. They were allowed to use chalk and they all used it. All participants completed the whole experimental session.

Climbing movements were recorded using an optoelectronic system (BTS SMART3D™ by BTS, Milan, Italy) with six infrared cameras (acquisition frequency 60 Hz), which sampled the locations of 12 retro-reflective markers. Coherent with the more traditional setup adopted for human movement analysis (for an extensive review see Cappozzo, Della Croce, Leardini, & Chiari, 2005; Chiari, Della Croce, Leardini, & Cappozzo, 2005; Della Croce, Leardini, Chiari, & Cappozzo, 2005; Leardini, Chiari, Della Croce, & Cappozzo, 2005), markers were attached in the most common repere points used in total body motion analysis when motion strategies are searched (Borghese, Bianchi, & Lacquaniti, 1996; Massion, 1992). In particular, markers were attached bilaterally at the acromion, the lateral epicondyle of the humerus, a point midway between ulna and radius epiphysis, the greater trochanter, the head of the fibula, the lateral malleolus, as shown in Fig. 2.

As a consequence, the body was modeled as a set of nine rigid segments (forearms, upper arms, trunk + head, thighs and legs) connected by spherical joints.

In general, markers were fixed directly on the climber's skin with adhesive tape except for the great trochanter markers, which were taped on the climber's clothes being careful to also firmly tape the clothes on the skin underneath. Markers visibility by the cameras was tested before acquisition.

The calibrated volume was $5.06 \times 3.41 \times 1.98 \text{ m}^3$ (X , Y and Z directions, respectively), where X direction is parallel to the climbing wall to the right, Y direction is vertical and Z

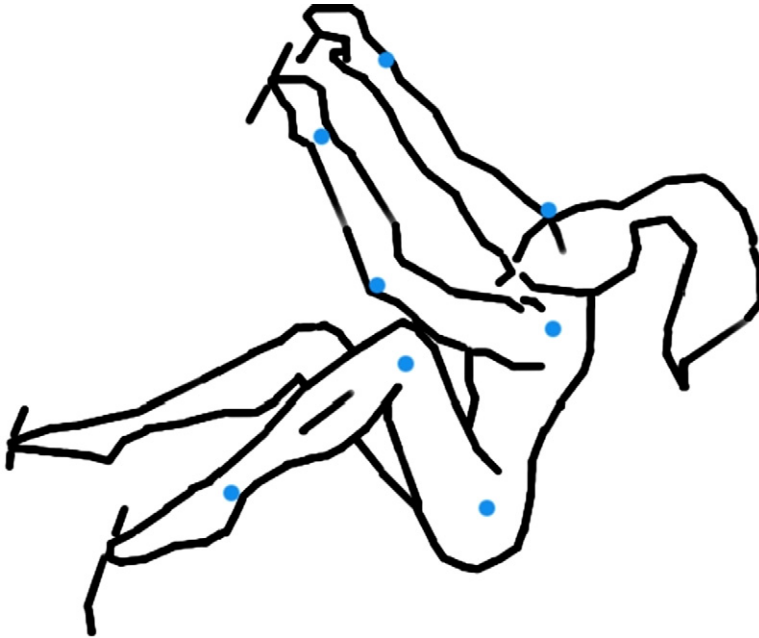


Fig. 2. Marker positioning: markers (blue dots) are positioned bilaterally at the acromion, the lateral epicondyle of the humerus, a point midway between ulna and radius epiphysis, the greater trochanter, the head of the fibula, the lateral malleolus. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

direction is horizontal, orthogonal to the climbing wall. An example of the reconstructed movement is given in Fig. 3, along with the calibrated volume (in green) and the computed body center of mass (magenta).

2.2. Data analysis

The three-dimensional coordinates of each marker were reconstructed for each trial. No correction was made to obtain the real joint center location from the marker coordinates. From these data the position of the center of mass (CM) was computed, frame by frame, as the weighted average of the position of the center of mass of the nine identified body segments

$$\text{CM} = \sum_{i=1}^9 P_i * \text{cm}_i,$$

where P_i is the percentage of mass of the i th segment with respect to the total body mass. The center of mass of the i th segment (cm_i) was computed from the marker coordinates, using the inertial parameters suggested by De Leva (1996). The contribution of the head was taken into account as summed to the trunk segment; the contribution of the feet and hands was neglected.

From the body CM coordinates (X, Y, Z), four different parameters were computed: the geometric entropy (H), the absolute velocity ($|V|$), the absolute acceleration ($|A|$) and the

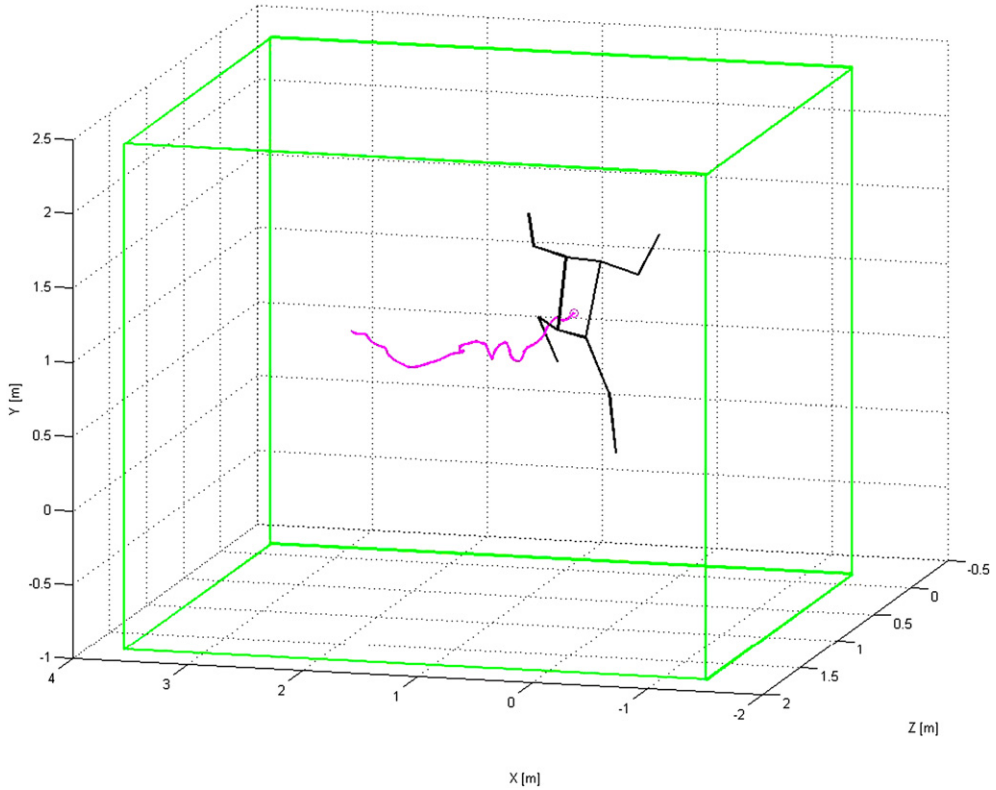


Fig. 3. Example of the reconstructed climbing movement; calibrated volume (green), body center of mass (magenta). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

power of acceleration time course (P); each parameter is described in detail in the following.

The geometric entropy (H) was computed by taking the natural logarithm of two times the length of the pattern travelled by the body center of mass (LP) divided by the perimeter of the convex hull around that path (c) (Cordier et al., 1993, 1994, see also Boschker & Bakker, 2002). H was computed on the frontal plane (XY), on the transverse plane (XZ) and on the sagittal plane (YZ) as

$$H = \ln \frac{2 * LP}{c},$$

where LP = CM path length, c = perimeter of the convex hull around LP.

In general, the entropy of a system is a measure of the amount of order vs. chaos of that system. As explained in Cordier et al. (1993, 1994) and Boschker and Bakker (2002), the geometric entropy reveals the amount of fluency/curvature of a curve: the higher the entropy, the higher the disorder of the system. In our case the geometric entropy H is associated with the oscillations of the climber's center of mass, represented by the length of its path. Thus, as also stated in Boschker and Bakker (2002), we would expect a low H value

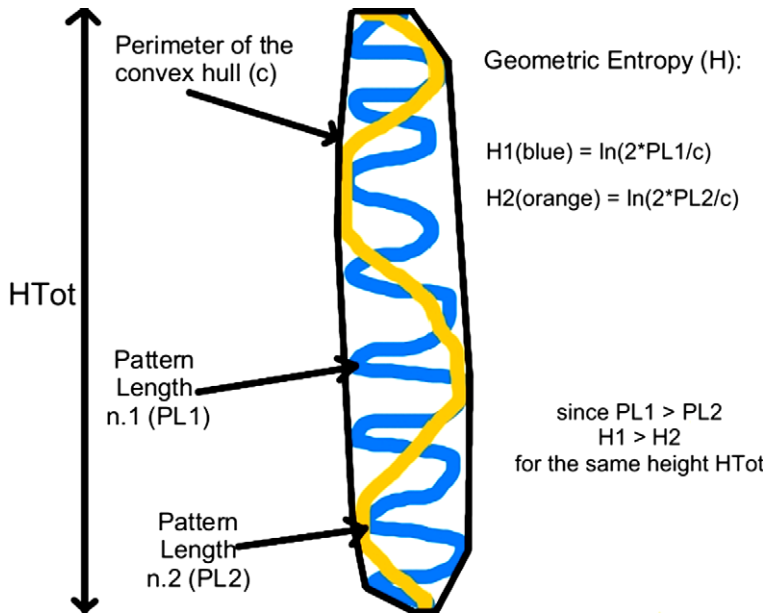


Fig. 4. Graphic explanation of the geometric entropy H .

(low chaos in the system) associated with a low energy expenditure and probably with a more effective climbing strategy. A graphic explanation of the geometric entropy H is given in Fig. 4 providing greater clarity.

The absolute velocity ($|V|$) is the magnitude of the velocity vector of the mass center and it is computed as

$$|V| = \sqrt{(\dot{X}_{CM}^2 + \dot{Y}_{CM}^2 + \dot{Z}_{CM}^2)}.$$

The velocity vector of the center of mass was computed from the center of mass trajectory by numerically deriving each trajectory component (X , Y and Z components).

Prior to the calculation of velocity and acceleration the values of the X , Y and Z coordinates of the center of mass were smoothed by a least-squares method using a Stavitsky and Golay filter (Stavitsky & Golay, 1964) as implemented in Matlab (Matlab, The Mathworks Inc., Natick, MA, USA) with a 140 m window as suggested by Willems, Cavagna, and Heglund (1995). Willems and colleagues used this filter for walking movements, but rock climbing movements seem to be in nearly the same, or even a lower, frequency range.

The absolute acceleration ($|A|$) is the magnitude of the acceleration vector of the mass center and it is computed as

$$|A| = \sqrt{(\ddot{X}_{CM}^2 + \ddot{Y}_{CM}^2 + \ddot{Z}_{CM}^2)}.$$

The acceleration vector of the center of mass was computed from the center of mass trajectory by numerically deriving twice each trajectory component (X , Y and Z components).

The power of acceleration time course (P) is the total power of the acceleration vector. It is computed according to signal processing theory as follows: the power spectral density

(PSD) is computed for each component of the acceleration vector, it is then integrated over all frequencies; the result is squared and the three components are added all together, giving

$$P = \sqrt{\left[\left(\int \text{PSD}(\ddot{X})df\right)^2 + \left(\int \text{PSD}(\ddot{Y})df\right)^2 + \left(\int \text{PSD}(\ddot{Z})df\right)^2\right]},$$

where PSD is the power spectral density of the acceleration. It describes how the acceleration power is distributed over the different frequencies. The PSD is computed as the Fourier transform of the autocorrelation function of the time series. The power of the acceleration signal is correlated with the actual physical power required to climb, as is clear from its mathematical definition (Bendat & Piersol, 1993).

At the end, based on the position and velocity of hands and feet markers, we measured the mean number of holds per acquired frame (lasting 1/60 s) used by each climber. The use of this index is justified by the fact that, in many climbing manuals (for example, Bjornstad, 1996; Reid & Falkenstein, 1992; Roper, 1971), it is recommended to use always at least three holds, moving just one limb at a time.

The correlation coefficient r was computed to find relationships between the computed parameters. Then, statistical significance of the correlation coefficient was tested using Student's t -test with threshold $p < .05$.

3. Results

Results are presented for the entire group of climbers as average values ($\pm$ standard deviation) and in greater detail for climber #4 and climber #6. These were at the edges of our group (as for the selected parameters) thus they clearly showed opposite climbing strategies.

Geometric entropy (H) evidenced interesting results in all three analyzed planes. In Table 1, the mean value ($\pm$ standard deviation) of the whole group and the mean values ($\pm$ standard deviation) for climber #4 and climber #6 are presented. In Fig. 5 a snapshot of the situation in all three planes is shown.

The differences in geometric entropy between climber #4 and climber #6 were all statistically significant: in the frontal and sagittal plane $p = .001$, while in the transverse plane $p = .033$. Comparison between climber #4 and the whole group also was statistically significant ($p < .05$) in all three planes, whereas comparison between climber #6 and the whole group was not.

The mean absolute velocity for the whole group was $|V| = 0.2 \pm 0.12$ m/s; climber #4 (0.17 ± 0.09 m/s) was the slowest of the group (and the most regular), whereas climber

Table 1
Geometric entropy (H) in frontal, sagittal and transverse plane

	Group	Climber #4	Climber #6
H (frontal)	0.35 ± 0.19	0.15 ± 0.11	0.41 ± 0.05
H (sagittal)	0.41 ± 0.21	0.19 ± 0.14	0.51 ± 0.06
H (transverse)	1.97 ± 0.51	1.39 ± 0.58	2.08 ± 0.17

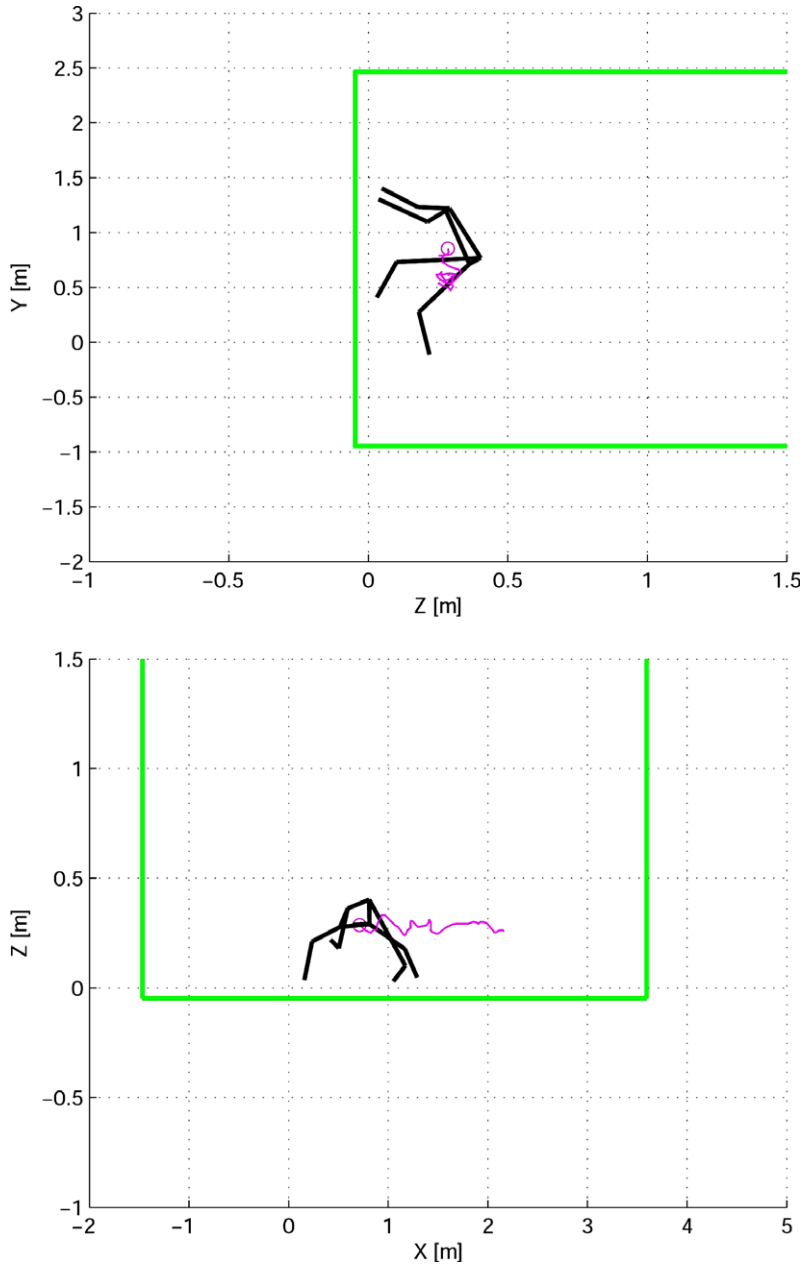


Fig. 5. Frontal, sagittal and transverse planes for the computation of the geometric entropy (H).

#6 was the fastest (0.24 ± 0.12 m/s). No statistically significant differences were found for this parameter ($p = .054$) between climber #4 and climber #6.

The mean absolute acceleration for the whole group was $|A| = 1.39 \pm 1.28$ m/s². At opposite sides we found climber #4 (1.02 ± 0.96 m/s²), who had the lowest acceleration

Table 2
Correlation coefficients

Parameter #1	Parameter #2	<i>r</i>	<i>r</i> ²	<i>p</i>
Mean holds per frame	Mean $ V $	-.87	76.18%	<.0001
Mean $ V $	Mean <i>P</i>	.76	52.67%	.0001
Mean $ A $	Mean <i>P</i>	.78	60.84%	<.0001

values and climber #6 showing higher and more sparse acceleration values ($1.56 \pm 1.67 \text{ m/s}^2$). The difference between climber #4 and climber #6 was statistically significant ($p = .019$).

The mean acceleration power *P* for the whole group was $0.34 \pm 0.17 \text{ m}^2/\text{s}^4$. Power values suggest that climber #4 moved on the wall using less power than all the other climbers (average value of $0.20 \pm 0.03 \text{ m}^2/\text{s}^4$), while climber #6 had the highest mean power value ($0.48 \pm 0.12 \text{ m}^2/\text{s}^4$) and also a higher standard deviation as compared to the other climbers. The difference between power values of climber #4 and climber #6 was statistically significant: $p = .012$.

In the whole group the mean number of holds per frame was 3.00 ± 0.22 ; 6 out of 12 climbers used, on average, more than 3 holds per frame in all their trials. Climber #4 used 3.05 ± 0.02 holds per frame, while climber #6 used 2.82 ± 0.19 holds per frame. The difference between climber #4 and climber #6 was statistically significant with $p < .001$.

Correlation coefficients *r*, percentage of variance in common between parameter #1 and parameter #2, *r*², and *p* values for interesting couples of parameters are shown in Table 2; the coefficients are computed taking into account the whole group of participants.

4. Discussion

As stated in the previous section, we will discuss the results focusing on the general behavior of the climbers' group, going into greater detail for the two climbers at the edges of the group (climber #4 and climber #6) to highlight the opposite climbing strategies they used.

The inverse correlation between the average number of holds per frame and the mean velocity ($|V|$) can be explained as follows: if the climber uses a smaller number of holds to move along the wall and up, he has to be quick enough to maintain equilibrium; conversely, if the number of holds is equal or greater than three, it is more likely that the participant climbs slowly, because his/her equilibrium is always under control. This principle is also very well known and explained by many authors of climbing technique books (Bjornstad, 1996; Peter, 2004; Reid & Falkenstein, 1992; Roper, 1971). Our group of 12 climbers fully respected this statement, with the average number of holds used in every frame being exactly three. Climber #4 and climber #6 showed opposite climbing styles. The first moved more slowly and carefully followed the 'three-holds-rule' as evidenced by the very low standard deviation values of both parameters: this means that climber #4 was constantly moving only one limb at a time, while the other three limbs were on the holds. On the other hand, climber #6 was quicker and used a smaller number of holds per frame than average; moreover, a greater standard deviation value for both parameters tells us that this climber probably moved inconstantly, sometimes snatching and sometimes staying still for thinking.

From the mathematical definition of geometric entropy (H) we can derive that the lower the H value, the lower the ‘disorder’ associated with the climber’s center of mass displacement, thus the higher the fluency of the climber’s movements. By comparing the H values in our group of climbers, we can notice that the difference between climber #4 and climber #6 remains large, the first being far below the group mean and the second being well above in all three planes considered. All the other climbers scored H values in between, evidencing that climber #4 was the most fluent and climber #6 was the most ‘chaotic’ of the group. If we agree that, in sport climbing, fluency can be a measure of effectiveness of the movement, our results agree with those of Boschker and Bakker (2002), who found that the lower the geometric entropy, the higher the climbing efficiency.

From Newton’s second law we can infer that the mean acceleration ($|A|$) is an index of the average global force used during the movement. It is difficult to compare our results with other studies, since we did not find any author who used an experimental setup similar to ours, nonetheless the average acceleration value of our group of climbers ($1.39 \pm 1.28 \text{ m/s}^2$) fully agrees with the average value given by Noè and colleagues ($1.4 \pm 0.5 \text{ m/s}^2$, horizontal acceleration) in their work about the role of anticipatory postural adjustments in ‘a rocking on heels’ task (Noè, Quaine, & Martin, 2004). As for the comparisons within our participants, again we found that climber #4 had an average acceleration value well below the group mean and a small standard deviation, while climber #6 showed the highest acceleration value and a higher standard deviation, if compared to the others. A small standard deviation value can be considered a sign of good repeatability: therefore the path followed by climber #4 was more consistent among the different trials than that followed by climber #6.

We did not find any work in the literature computing power in rock climbing, thus our discussion was limited to the analysis of our group of climbers. We noticed that power values confirm the results of the acceleration time course. Moreover, these two parameter showed a good linear relationship as evidenced by the correlation coefficient r . The comparison of the power values within our 12 climbers points out again that climber #4 and climber #6 used opposite movement strategies: climber #4 moved with more agility and technique than power, the opposite for climber #6, who used more power and force than agility and technique.

For all parameters considered, standard deviation values were generally lower for climber #4 compared to all other climbers of the group: this result can be interpreted as a higher degree of consistency in repeating the same motion; it could also be linked to a better control of the body motion.

Qualitative visual analysis of the climbing styles adopted by climbers #4 and climbers #6 during the experimental session confirms the numerical results reported here. Climber #4 moved naturally on the rock wall, linking together a series of fluid movements; on the other hand, climber #6 executed a set of quick movements, using snatches and force to advance along the rock wall.

5. Conclusions

This pilot study identified the motion of the body center of mass in rock climbing in a group of recreational climbers. Our main aim was to search for common patterns and for different climbing strategies, by means of kinematics analysis of a simple climbing task.

The evaluated parameters demonstrate the existence of two major strategies: the first is based on ‘agility’ and requires less speed and less power for the movement to be successfully completed, while the second is based on ‘power’ and requires more speed and more force for the movement to be successfully completed. We think that the first strategy could be the more effective, because it requires less power, more fluency and more control of the equilibrium, as also stated in many climbing reference books (Fyffe & Peter, 1997; Peter, 2004).

In sum, our results suggest that a good climber minimizes power during the whole movement. This is well captured by a low value of geometric entropy; this global variable gives also a quantitative evaluation of the smoothness of the center of mass motion, which well reflects the fluency of the motion. Our data do not permit to conclude that this way of climbing represents a gold standard strategy, however this hypothesis will be a starting point for future experiments. In any case, all the computed parameters could be used for future comparisons with different groups of climbers (elite athletes, beginners, children, for instance).

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